

SCQ (Single Correct Type) :

- In context with the highly pure dilute solution of sodium in liquid ammonia, which of the following is the incorrect statement ?
 (A) In dilute solution the main species are metal ions (Na^+) and electrons, which are solvated (i.e. ammoniated)
 (B) It is paramagnetic, with approximately one unpaired electron per metal atom.
 (C) It acts as powerful oxidizing agent.
 (D) It shows blue color and exhibits electrical conductivity.
- A vessel contains two gases A and B in the mass ratio 2 : 1 respectively. If the above gas mixture is allowed to diffuse through a hole, the molar composition of the mixture coming initially out of the hole is 1 : 4. Find the simplest whole number ratio of the moles of gases B and A present in the vessel in the beginning respectively.
 (A) 0.5 (B) 1.0 (C) 1.5 (D) 2.0
- $\text{FeS} + \text{H}_2\text{SO}_4 \longrightarrow \text{B} \uparrow$, which is incorrect about gas 'B' ?
 (A) It produce white ppt. with FeCl_3 .
 (B) It produce black ppt. with $(\text{CH}_3\text{COO})_2\text{Pb}$.
 (C) It decolorise the purple colour of KMnO_4 .
 (D) It produce purple colour solution with sodium nitroprusside.
- Which of the following reactions of potassium superoxide supply oxygen gas in the breathing equipment used in space capsules and submarines ?
 (1) Reaction of superoxide with nitrogen in the exhaled air
 (2) Reaction of superoxide with moisture in the exhaled air
 (3) Reaction of superoxide with carbon dioxide in the exhaled air
 (A) (1), (2) and (3) (B) (2) and (3) only (C) (2) only (D) (1) and (2) only
 (E) None of these

- Salt mixture** $\xrightarrow{\text{dil. HCl}}$ **white pp.** $\xrightarrow[\text{and filtered under hot condition}]{\text{Heated}}$

$\xrightarrow{\text{Cooled}}$ Filtrate $\rightarrow$ white needle like crystal
 $\xrightarrow{\text{NH}_3 \text{ Sol.}}$ Residue $\rightarrow$ Clear solution

(A) Pb^{2+} and Hg_2^{2+} (B) Ag^+ and Hg_2^{2+} (C) Pb^{2+} and Ag^+ (D) None of these

MCQ (Multiple Correct Type) :

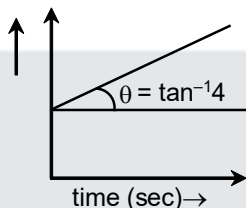
- In which of the following, the elevation in boiling point of the final solution is more than that of initially taken solution (No one is limiting reagent) [Assume $\alpha = 1$ where ever needed]
 (A) $\text{K}_4[\text{Fe}(\text{CN})_6] (\text{aq}) + 2\text{CuSO}_4(\text{aq}) (\text{added}) \longrightarrow \text{Cu}_2[\text{Fe}(\text{CN})_6] \downarrow + 2\text{K}_2\text{SO}_4(\text{aq})$

- (B) $\text{Al}_2(\text{SO}_4)_3(\text{aq}) + 3\text{BaCl}_2(\text{aq}) \text{ (added)} \longrightarrow 3\text{BaSO}_4\downarrow + 2\text{AlCl}_3(\text{aq})$
- (C) $\text{AgNO}_3(\text{aq}) + 2\text{KCN}(\text{aq}) \text{ (added)} \longrightarrow \text{K}[\text{Ag}(\text{CN})_2](\text{aq}) + \text{KNO}_3(\text{aq})$
- (D) $4\text{KI}(\text{aq}) + \text{HgCl}_2(\text{s}) \text{ (added)} \longrightarrow \text{K}_2[\text{HgI}_4](\text{aq}) + 2\text{KCl}(\text{aq})$
7. Which of the following option(s) is/are correct for solution of liquid NH_3 with alkali metal?
- (A) It is paramagnetic (B) It is Blue in colour
(C) It is good conductor of electricity (D) It is non conductor of electricity
8. A salt mixture contain 0.018 mol each of NaI, NaBr and NaCl. When the mixture is dissolved in 200 mL of 0.24 M AgNO_3 , some precipitates are formed. Volume of final solution is 200 mL. Given (K_{sp} of AgI, AgBr and AgCl = 10^{-17} , 10^{-13} , 10^{-10} respectively). Which of the following is/are correct?
- (A) Nearly all I^- and Br^- are precipitated out.
(B) $[\text{Ag}^+]$ in the final solution is $\frac{10^{-8}}{9}$ M
(C) Total moles of silver salt in the precipitate is 0.036 mol.
(D) Concentration of $[\text{I}^-]$ and $[\text{Br}^-]$ in the final solution is nearly 3×10^{-9} M and 3×10^{-5} M respectively.
9. Aniline, $\text{C}_6\text{H}_5\text{NH}_2$ react with water according to the equation

$$\text{C}_6\text{H}_5\text{NH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{C}_6\text{H}_5\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$$
In a 0.180 M aqueous aniline solution the $[\text{OH}^-] = 8.80 \times 10^{-6}\text{M}$
The value of the base ionization constant K_b for $\text{C}_6\text{H}_5\text{NH}_2(\text{aq})$ and the percent ionization of $\text{C}_6\text{H}_5\text{NH}_2$ in this solution are :
- (A) 4.3×10^{-10} (B) 3.1×10^{-10} (C) $4.9 \times 10^{-3}\%$ (D) $2.4 \times 10^{-3}\%$
10. Amongst the following select correct statement(s) :
- (A) Stability of lyophilic colloids is mainly due to the strong interaction between dispersed particle and dispersion medium
(B) Gelatin has considerably low value of gold number and is effective protective colloid
(C) Zeta potential is also responsible for stability of lyophobic colloid solution
(D) Surface tension of lyophilic colloidal solution is lesser than that of dispersion medium
11. In a crystalline solid, B^{2-} ions form cubic close packing and A^{+2} ions occupy appropriate voids. If radius of A^{+2} ion is 150 pm and that of B^{2-} ion is 450 pm, then ($\sqrt{3} = 1.73$) :
- (A) Coordination number of cation is 4
(B) Coordination number of anion is 4
(C) 66.66% of 3D (TV+OV) voids are unoccupied
(D) Edge length of unit cell is 1387.28 pm
12. An ideal gas at initial pressure P_i and volume V_i undergoes reversible expansion to the same volume V_f either by isothermally or adiabatically. Which of the following statement(s) is /are correct (where the symbols have their usual meaning) ?
- (A) $P_f(\text{adiabatic}) < P_f(\text{isothermal})$ (B) $|W(\text{adiabatic})| < |W(\text{isothermal})|$
(C) $T_f(\text{adiabatic}) < T_f(\text{isothermal})$ (D) $q(\text{adiabatic}) < q(\text{isothermal})$

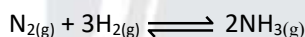
13. Which of the following statement(s) is/are correct ?
- (A) Entropy of a gas undergoing adiabatic free expansion remains constant because heat exchange in adiabatic process is zero
- (B) ΔG for the change, $\text{H}_2\text{O} (\ell, 0.9 \text{ atm}, 373 \text{ K}) \rightarrow \text{H}_2\text{O} (\text{g}, 0.9 \text{ atm}, 373 \text{ K})$ is negative
- (C) Electrical work is non – PV work
- (D) $\Delta G = \Delta H - T\Delta S$ is always applicable for deciding spontaneity of a process in closed system

14. For the reaction : $2\text{C}_2\text{F}_4 \longrightarrow \text{C}_4\text{F}_8$

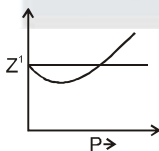


Then

- (A) Order of reaction is 3.
- (B) Half life is $\frac{3}{4}$ min if initially 1 M C_2F_4 is taken.
- (C) Reaction must be complex.
- (D) Disappearance constant w.r.t C_2F_4 is 4 min^{-1}
15. Which of the following statement is/are correct regarding real gas ?
- (A) In the vander Waals equation, 'a' signifies intermolecular attraction.
- (B) At critical point attractive tendency dominate.
- (C) Below Boyle temperature gas shows negative deviation at all range of pressure.
- (D) Boyle temperature of a gas depends upon its nature.
16. Which of the following is/are true for the equilibrium constant (K_{eq}), forward rate constant (k_f) and backward rate constant (k_b) of the given exothermic reaction ?



- (A) K_{eq} will decrease on increasing temperature
- (B) k_f will decrease on increasing temperature
- (C) k_b will increase on increasing temperature
- (D) K_{eq} will increase on increasing temperature
17. A gas shows following graph at 25°C .

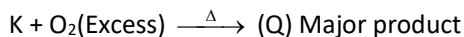
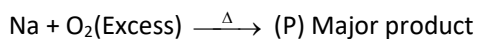


Which is/are correct for the gas ?

- (A) Gas shows only negative deviation throughout the graph
- (B) Boyle temperature of gas must be more than 25°C .
- (C) Critical temperature of gas must be more than 25°C .
- (D) In the high pressure region, the gas is less compressible than ideal gas at 25°C .

(Numerical Value type) :

18. Consider the following chemical reactions:



Determine the value of expression $5(x + 2y)$ (where x and y are atomicity in compound (P) and (Q) respectively).

19. Equal volume of 0.2 M CH_3COOH is mixed with 0.2 M NaOH solution then determine pH of resulting solution? Write your answer after multiplying by 9. [$K_a(\text{CH}_3\text{COOH}) = 10^{-5}$]

20. In the process $PT = \text{constant}$, an ideal gas has molar heat capacity $C = 37.35 \text{ J mol}^{-1}\text{K}^{-1}$. Find the total number of degree of freedom of molecules in the gas. ($R = 8.3 \text{ J/mol K}$)

21. 0.01 mol of an organic compound ($\text{C}_x\text{H}_y\text{O}_x$) was treated with 224 ml of $\text{O}_2(\text{g})$ at 1 atm and 273 K. After combustion the total volume of gas was 560 ml at 1 atm and 273 K. On treatment with KOH solution the volume decreases to 112 ml. What is the value of 'x'?

22. It is observed that functioning of neuronal circuits in our brain is driven by the energy available from combustion of glucose. Calculate the amount of glucose (in gm) which should be burnt per hour to produce sufficient energy for brain which operates at $\frac{128}{9}$ watts.

Given : ΔH at 300 K for combustion of Glucose = -3000 KJ/mol

ΔS at 300 K for combustion of Glucose = 180 J/mol K

23. If Aufbau rule is not obeyed and instead electronic filling occurs orbit by orbit till saturation is reached. The percentage change in sum of all $(n + l)$ values for unpaired electrons in an atom of iron is 10 x. What is the value of x?

24. Consider the following compounds:

(i) ICl (ii) IBr (iii) ICl_3 (iv) SF_4

(v) BrF_3 (vi) IF_3 (vii) BrF_5 (viii) IF_7

Calculate the value of $(P - S)$.

P : Total number of polar compounds

S : Total number of non planar compounds

25. Consider, the following compounds in solid state:

(i) N_2O_5 (ii) Cl_2O_6 (iii) PCl_5 (iv) I_2Cl_6

How many of the cationic and anionic part of above compounds are sp^3 hybridized.

Answer key

1.	(C)	2.	(D)	3.	(D)	4.	(B)	5.	(C)
6.	(ABC)	7.	(ABC)	8.	(AD)	9.	(AC)	10.	(ABCD)
11.	(ABCD)	12.	(ABCD)	13.	(BC)	14.	(AC)	15.	(A,B,D)
16.	(A,B, C)	17.	(B, D)	18.	(50.00)	19.	(81.00)	20.	(5)
21.	(4)	22.	03.00	23.	05.00	24.	04.00	25.	3

